

Multi-mode Traffic Flow State Estimation Based Adaptive Signal Phase Decision and Control Mechanism for Service-oriented C-ITS

Junle Zhou¹, Dagang Shen¹, Yechen Ren¹, Yaxuan Xie¹, Kailong Zhang^{1*}, Thi-Mai-Trang Nguyen²

Abstract—Traditional traffic signals employ the fixed-time control mechanism, which is incapable of adapting to the dynamic changes in traffic conditions and cannot provide more reasonable scheduling for vehicles of different service urgencies, such as firefighting and public security. This significantly impacts the traffic efficiency and quality at intersections. In response to these issues, this paper studies and proposes an adaptive traffic signal phase decision and control framework based on multi-mode traffic flow state estimation. Initially, we constructed networked vehicle model with service priority attributes, as well as models for traffic objects such as roads and intersections. Subsequently, we researched and proposed a multi-mode, dynamic estimation method for traffic flow situations that integrates vision, communication, and multi-priority. Building upon this foundation, we designed and implemented a signal phase decision and control method that combines logical rules and deep reinforcement learning to effectively handle various intersection scenarios. Finally, we developed a prototype intelligent signal controller system and integrated a semi-physical simulation environment using SUMO, CARLA, and other tools. We comprehensively validated the designed methods based on typical scenarios.

Index Terms—Traffic Signal Control; Multi-mode Estimation; Adaptive Control Strategy; Prototype System Development

I. INTRODUCTION

As an important component of road networks, the fluidity of traffic flow at intersections is a key factor affecting urban traffic efficiency. Among various measures to alleviate traffic congestion, optimizing traffic signal control strategies at intersections is one of the quickest and most effective methods [1]. Traffic signal control methods can generally be categorized into Fixed-time methods, Experience-driven methods and Adaptive methods. Fixed-time control methods control signal lights based on predetermined time intervals, without dynamic adjustments. Experience-driven methods use historical data to adjust timing plans but lack flexibility and efficiency in real-time dynamic traffic flows. Adaptive methods like SCOOT [2] and SCATS [3] utilize specific rules or models to optimize signal timing parameters and adjust signal lights based on real-time traffic conditions. However, these methods are

constrained by fixed environments, have limited responsiveness to traffic behavior, and cannot adaptively schedule based on different traffic flow patterns or prioritize emergency vehicle dispatch.

With the development of computer and information technology, transportation is evolving towards the direction of Cooperative Intelligent Transportation Systems (C-ITS), which has rendered the realization of deep collaboration between vehicles, roads, and the cloud and the improvement of overall efficiency through intelligent control of individual traffic entities. In the field of C-ITS, the increasingly integrated application of technologies such as artificial intelligence, computer vision, vehicular networking, and big data provides strong support for effective solutions to traffic problems. For example, real-time sensation and estimation technology of traffic flow state in each lane at intersections based on computer vision have laid a solid foundation for adaptive signal management and control. Additionally, leveraging Vehicle-to-Everything (V2X) communication technology enables online perception of the emergency level of vehicles, making it possible to prioritize the dispatch of emergency vehicles.

In this context, the metric of Quality of Service (QoS) has increasingly gained prominence, necessitating that the control actions of traffic systems be subservient to the service attributes of vehicles. This imperative has catalyzed the evolution of Cooperative Intelligent Transportation Systems (C-ITS) towards the Service-oriented Cooperative Intelligent Transportation System (SoC-ITS) paradigm [4].

To this end, this study focuses on the intelligent scheduling of intersection traffic flow within the framework of Service-oriented C-ITS, and investigates and designs an adaptive traffic signal phase and timing decision and control framework, as well as its core control methodologies, based on multi-mode traffic flow state estimation. The key contributions of our work are as follows:

- We design and propose an adaptive traffic signal phase decision and control framework based on multi-mode traffic flow state estimation.
- Based on this framework, we propose a multi-mode traffic flow state estimation method based on computer vision and v2x technologies as well as an adaptive traffic signal phase decision and control method composed of deep reinforcement learning and logical rules.
- We design and construct a prototype system utilizing a complete set of hardware and software configurations and validate framework on this system.

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The remainder of this paper is organized as follows: We first analysis the current related work in Section 2. The modeling of the traffic scenario is conducted in Section 3. We then describe our multi-mode traffic flow state estimation and adaptive traffic signal phase decision and control method in Section 4 and Section 5 respectively. The system design, implementation, and experimental work are elucidated in Section 6. Finally, we provide a summary of this paper in Section 7.

II. RELATED WORK

With the extensive application of information technology in the field of intelligent transportation, it has become feasible to perceive the status of intersections and roads. This has led to the emergence of numerous promising solutions for the Traffic Signal Control (TSC) problem and the research of adaptive control method has become a hot topic among them.

Overall, research in adaptive traffic signal control can be broadly classified into model based, intelligent computation based, and reinforcement learning based control approaches. For instance, Cao et al. [5] designed a method to detect potential vehicle queue overflows and defined three traffic light transition triggers based on the detection of queue overflows to prevent traffic congestion at intersections. Song et al. [6], aiming to reduce emissions on main roads, integrated a microscopic traffic flow model with a vehicle emission model and employed an Improved Simulated Annealing Genetic Algorithm (ISAGA) to solve for the optimal signal control scheme. Kar et al. [7] considered parameters such as the number of vehicles, average speed, and average distance between vehicles per lane, implementing a low-cost intelligent traffic signal control method through the Firefly Algorithm. Li et al. [8] mapped the number of queuing vehicles to a low-dimensional action space and designed a Q-network with an autoencoder to learn optimal control method for a single intersection. Zeng et al. [9] constructed a recurrent neural network to process state sequences and used experience replay and a recursive Q-network for training to address the problem of traffic signal control. The above research lays the foundation for the research work presented in this paper.

III. MODELING TRAFFIC OBJECTS

The objects of traffic signal scheduling are the participants in traffic at intersections and within a certain surrounding area. Therefore, we have modeled the relevant objects.

A. Vehicle Model

Vehicles are the core and the subject of scheduling in cooperative traffic signal services. We have endowed it with various attributes to reflect its state and abstracted it into the following eq. 1.

$$V = \langle id, pri, pos, speed, travel, wait, lane_{id} \rangle \quad (1)$$

The definitions of each attribute in eq. 1 are shown in Table I.

TABLE I. DEFINITIONS OF VEHICLE MODEL'S ATTRIBUTES

Attribute	Definition
$v.id$	id
$v.pri$	priority weight
$v.pos$	location
$v.speed$	speed
$v.travel$	travel time
$v.wait$	waiting time
$v.lane_{id}$	associated lane id

B. Road Model

Typically, roads are composed of one or more lanes. This paper primarily focuses on the lanes that approach intersections from the roads connected to the intersection and disregards other opposite lanes. Turn types of lanes typically encompass only three basic categories including left type, straight type, and right type, as well as compound turn types composed of these three basic turn types.

Therefore, we define a road as a group of turn type objects, denoted by R , and define turn type object as a collection of

lanes, denoted by T . Then we represent lanes with L , so the road and turn type object (TTO) can be defined as eq. 2 and eq. 3 respectively.

$$R = \{T_1, T_2, \dots, T_m\} \quad (2)$$

$$T = \{L_1, L_2, \dots, L_n\} \quad (3)$$

In these equations, parameter m is the number of basic turn types on a given road and n is the number of lanes which have this turn type (Lane with compound turn types can belong to multiple TTOs at the same time). The attributes of lane L are defined as eq. 4, with each attribute and its definition detailed in Table II.

$$L = \langle id, area, types, n_{veh}, qlen_{veh}, wait_{veh}, fwait_{veh}, pri_{veh}, road_{id}, phases_{id} \rangle \quad (4)$$

Within these attributes, $l.qlen_{veh}$ refers to the number of vehicles in the stopped state, while $l.n_{veh}$ refers to the number of all vehicles within the range of lane. Due to the fact that the waiting time of the first vehicle is often the longest, we utilize the waiting time of the first vehicle ($l.fwait_{veh}$) as the representative for the maximum waiting time.

The attributes of TTO are defined as eq. 5, with each attribute and its definition detailed in Table III.

$$T = \langle id, type, n_{veh,max}, qlen_{veh,max}, wwait_{veh}, fwait_{veh}, pri_{veh}, phases_{id} \rangle \quad (5)$$

The attribute with the subscript "max" in Table III refers to the maximum value of the corresponding attribute for all lanes belonging to that TTO. The calculate method of $t.wwait_{veh}$ is eq. 6. Q in this equation represents the set of all vehicles belonging to this TTO.

$$t.wwait_{veh} = \sum_{v \in Q} v.wait \cdot v.pri \quad (6)$$

TABLE II. DEFINITIONS OF LANE MODEL'S ATTRIBUTES

Attribute	Definition
$l.id$	id
$l.area$	area coordinate
$l.types$	turn type set
$l.n_{veh}$	vehicle num
$l.qlen_{veh}$	waiting quene length
$l.wait_{veh}$	vehicle waiting time sum
$l.fwait_{veh}$	max vehicle waiting time
$l.pri_{veh}$	sum of priority weights
$l.road_{id}$	associated road id
$l.phases_{id}$	associated phase id set

TABLE III. DEFINITIONS OF TTO MODEL'S ATTRIBUTES

Attribute	Definition
$t.id$	id
$t.type$	turn type
$t.n_{veh,max}$	max vehicle num
$t.qlen_{veh,max}$	max waiting quene length
$t.wwait_{veh}$	weighted waiting time sum
$t.fwait_{veh}$	max vehicle waiting time
$t.pri_{veh}$	sum of priority weights
$t.phases_{id}$	associated phase id set

Furthermore, since vehicles making right turns generally do not conflict with other vehicles, the mechanism described in this paper does not manage right-turning vehicles.

C. Intersection Model

As the basic unit of traffic signal phase control, many intersection's attributes affect the signal control process significantly.

In this paper, the attributes of the intersection object are abstracted into eq. 7, in which the attributes and their specific definitions are shown in Table IV.

$$J = \langle id, ps, phase_c, pt_{min}, rs, ped, area_{ped}, mwait_{ped}, pass_{ped}, pn_{veh}, tt_{veh} \rangle \quad (7)$$

IV. MULTI-MODE TRAFFIC FLOW STATE ESTIMATION MECHANISM DESIGN

Achieving effective scheduling outcomes necessitates accurate, rapid and comprehensive traffic situational awareness capabilities. Therefore, in response to the characteristics of traffic scenarios, we have designed a multi-mode traffic flow state estimation mechanism based on computer vision technology and Vehicle-to-Everything (V2X) communication. This mechanism leverages computer vision technology to collect information on various types of ordinary low-priority vehicles. Meanwhile, V2X

TABLE IV. DEFINITIONS OF INTERSECTION MODEL'S ATTRIBUTES

Attribute	Definition
$j.id$	id
$j.ps$	phase set
$j.phase_c$	current phase
$j.pt_{min}$	min phase time
$j.rs$	connected road set
$j.ped$	whether has pedestrian
$j.area_{ped}$	pedestrian area coordinate
$j.mwait_{ped}$	max pedestrian waiting time
$j.pass_{ped}$	min pedestrian pass time
$j.pn_{veh}$	passed vehicle num
$j.tt_{veh}$	total travel time of passed vehicle

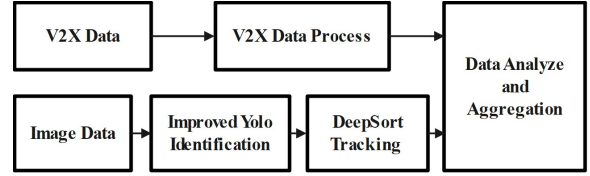


Figure 1. Multi-Mode Traffic Flow State Estimation Process

communication technology is utilized to gather information of high-priority vehicles (such as emergency vehicles). Ultimately, all collected information is synthesized. The process is illustrated in Fig. 1.

A. Computer Vision Technology based Estimation Method

YoloV5: YoloV5s algorithmic model primarily consists of four components: the Input, the Backbone, the Neck, and the Head detection network. The Input stage employs data augmentation techniques, which include random cropping, scaling, and the blending of features with labels for enhanced detection capabilities.

The Backbone serves as the primary network for feature extraction from the image. In this study, a C3 bottleneck architecture is implemented within the Backbone, encompassing a Cross Stage Partial (CSP) [10] structure, Bottleneck blocks, and an SPPF (Spatial Pyramid Pooling) module designed to address the issue of repetitive feature extraction.

The Neck structure is tasked with the fusion and integration of features. Its architecture combines the FPN (Feature Pyramid Network) [11] and PAN (Path Aggregation Network) [12] to form a PAFPN (Parallel and Asymmetric Feature Pyramid Network).

To enhance the receptive field and improve the feature extraction for distant and occluded portions of vehicles, this study employs the CARAFE (Content-Aware ReAssembly of FEatures) [13] upsampling operator as the substitute for the original nearest-neighbor interpolation upsampling. This operator generates more refined and accurate upsampling results by calculating weights and performing a weighted average, taking into account the local contextual information. The principle of it can be expressed as eq. 8 and eq. 9.

$$w_{ij,m,n} = \frac{\exp(g_{ij,m,m})}{\sum_{p,q} \exp(g_{ij,p,q})} \quad (8)$$

$$g_{ij,m,n} = \frac{1}{K} \sum_{p,q} F_{ij} \times T_{mn,pq} \quad (9)$$

In these given equations, $w_{ij,m,n}$ denotes the weight of the pixel value at location (i, j) at the position (m, n) , $g_{ij,m,n}$ is an intermediate variable for the weight calculation, K is a hyperparameter, F_{ij} represents the pixel value at location (i, j) of the input feature map F , and $T_{mn,pq}$ is a learnable transformation matrix.

After obtaining the features of the FPN, the Head network is responsible for performing the detection task and outputting the detected vehicle positions. In order to enhance the

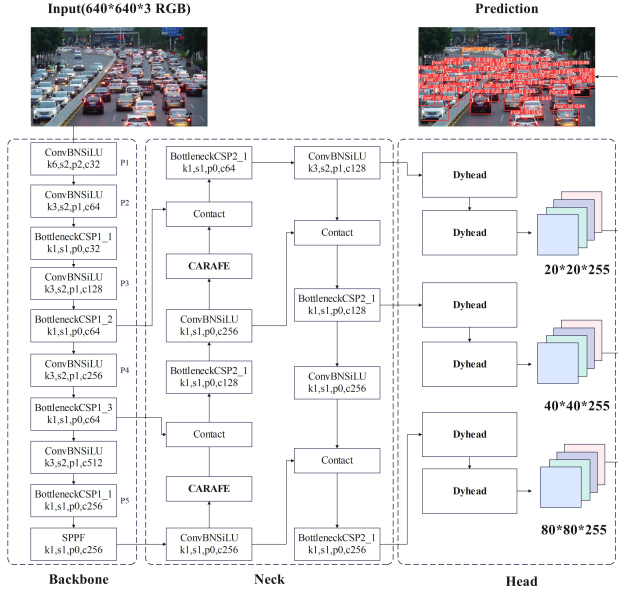


Figure 2. Structure of Improved Yolov5s Network

effectiveness of vehicle detection at different distances, we employ Dyhead [14] as the detection head, which is based on dynamically learning and adjusting the head structure within the network to accommodate the feature representation and prediction requirements of different targets. The structure of improved Yolov5s network is shown in Fig. 2.

DeepSort: DeepSort algorithm is tasked with tracking vehicles. As an improved version of the Sort tracking algorithm, DeepSort employs a trajectory processing and Kalman filtering framework that is largely identical to that of Sort. However, it introduces the motion and appearance information of the targets as association metrics and incorporates a cascade matching strategy, which significantly improves tracking effectiveness.

DeepSort algorithm utilizes a Kalman filter to predict the likely position of the tracked targets in the subsequent frame. For each tracked target k , DeepSort keeps a record of the number of frames that have elapsed since the last successful match up to the current frame. Tracking for that target is discontinued when this count exceeds a predetermined threshold. And for these newly emerging potential targets, if they are successfully matched for the next several frames (we set it to 2 frames), it is trusted to be a new tracking target. Otherwise, it is deemed unnecessary to continue tracking, and its features are discarded.

The frame-by-frame matching algorithm of DeepSort encompasses two metrics: the Mahalanobis distance between the predicted and actual bounding boxes, and the minimum cosine distance between the current target's feature vector and the vectors recorded in the historical feature database, which are presented as eq. 10 and eq. 11. [15]

$$d^{(1)}(i, j) = (d_j - y_i)^T S_i^{-1} (d_j - y_i)^T \quad (10)$$

$$d^{(2)}(i, j) = \min \{1 - r_j^T r_k^{(i)} | r_k^{(i)} \in R_i\} \quad (11)$$

where we denote the projection of the i -th track distribution into measurement space by (y_i, S_i) and the j -th boundingbox detection by d_j ; r_j represent appearance descriptor. By integrating the two with a weighted fusion, eq. 12 is obtained.

$$c_{ij} = \lambda d^{(1)}(i, j) + (1 - \lambda) d^{(2)}(i, j) \quad (12)$$

To mitigate the issue of deteriorated prediction accuracy in Kalman filters due to target occlusions or extended periods of frame loss, DeepSort also incorporates a cascade matching strategy to ensure that targets can be matched before they are occluded or lost. With these solutions in place, DeepSort is capable of effectively accomplishing tracking tasks.

Recognition Result Process: After acquiring the identification and tracking results, the vehicle priority is initially set to a low priority by default. Subsequently, a coordinate transformation algorithm in paper [16] is utilized to convert the central coordinates of the vehicle's recognition bounding box into real-world coordinates.

B. V2X Based High-Priority Vehicle Perception Method

Compared to computer vision recognition technology, V2X communication technology is more stable and capable of accurately transmitting vehicle information. Therefore, it is highly suitable for the transmission of high-priority vehicle information using this technology.

When a high-priority vehicle is on a mission, it is required to utilize a V2X module to signal the intersection controller as it approaches the intersection. It must report its priority, coordinate location, and turning type, among other data, and then await scheduling by the intersection controller.

C. Data Analyze and Aggregation

Upon receiving data from vehicles of all priorities, the location reported by the emergency vehicle is compared with the vehicle coordinates perceived through computer vision to ascertain the corresponding tracking ID for that high-priority vehicle. The attributes obtained are then used to update the information for the vehicle associated with that ID, thus integrating the data acquired from both sensing methods.

The lane attributes are derived from a comparison between the vehicle's coordinates and the road lane boundaries. The velocity is then computed based on the time difference between adjacent frames and the distance the vehicle has moved. Consequently, the number of waiting time can be calculated. The travel time attribute is obtained based on the time difference between the vehicle's entry and exit from the field of view. This provides us with information about each vehicle, which is represented by eq. 1.

With knowledge of the lane attributes, the vehicle's associated turning type and phase can also be identified (since high-priority vehicles report their turning type, they are only counted under one turning type, even if they are located in a compound lane).

After successfully acquiring data from various types of vehicles, integrating this data by lane, TTO, and intersection

yields the required traffic flow situation data, as represented by eq. 4, eq. 5, and eq. 7.

V. ADAPTIVE SIGNAL PHASE TIME DECISION AND CONTROL METHOD

To ensure the versatility and scheduling efficiency of the system, our proposed framework is composed of two types of scheduling algorithms: Deep Reinforcement Learning (DRL)-based signal phase decision and control algorithm, and logical rule-based signal phase decision and control algorithm. Among them, the DRL scheduling algorithm offers the best scheduling performance, it provides the most efficient strategy generation and minimizes travel time and waiting time. However it has specific requirements for the intersection scenarios in which it is deployed, and the phase schemes it utilizes, achieving maximum efficiency only in scenarios that have been specifically optimized/trained for. The logical rule-based algorithm, while slightly less effective than the reinforcement learning scheduling algorithm, is more universally applicable to various types of intersections, offering better generality.

In environment of reality, one of the significant factors affecting the vehicular traffic efficiency at intersections is pedestrians crossing the street. Therefore, to improve the practicality, we have integrated a pedestrian detection mechanism. Specifically, the system employs computer vision technology to detect the pedestrian waiting areas at both ends of the zebra crossing. When a pedestrian who appears to be waiting to cross the intersection is detected, the system mandates the allocation of a phase within the maximum waiting time that will enable the pedestrian to cross the intersection.

A. Logical Rule-Based Signal Phase Decision and Control Algorithm

The process of this method including two parts, which is shown in Fig. 3. PW and TP in Fig 3 represent phase weight and target phase respectively.

The phase weight is calculated with eq. 13.

$$PW_{p_i} = \sigma_1 \sum_{l \in LS_i} l.n_{veh} + \sigma_2 * \max(l.fwait_{veh} \mid l \in LS_i)^\tau + \sigma_3 * \sum_{l \in LS_i} l.pri_{veh} + \sigma_4 * (t_c - t_{l,i}) \quad (13)$$

In eq. 13, LS_i represents the set of lanes belonging to the i -th phase. The first term denotes the total number of vehicles belonging to the current phase, serving as a measure of traffic pressure; the second term is the maximum vehicle waiting time for this phase, which is used to prevent the occurrence of vehicles experiencing prolonged waiting times. Additionally, the exponent τ ($\tau > 1$) ensures that as the maximum waiting time increases, the weighting of this term accelerates more rapidly; the third term is the sum of the priority weights of all vehicles in the current phase, which is utilized to implement a service-oriented priority scheduling mechanism; the fourth term represents the time difference between the current time and the last occurrence time of this phase, which prevents misjudgment of vehicle turning types in compound lanes during scheduling, thus avoiding situations where vehicles are unable to proceed for extended

periods. σ_1 to σ_4 are coefficients used to adjust the proportion of each term.

After acquiring PW of all phases, we select the most weighted phase as TP . After getting TP , we use eq. 14 to calculate its duration.

$$T_{p_i} = \rho_1 * \max(l.n_{veh} \mid l \in LS_i) * k_i \quad (14)$$

In the given eq. 14, parameter i represents id of TP , k_i represents the traffic flow trend coefficient for the target phase. Its initial value is set to 1. When it is consistently observed that the maximum number of vehicles of all lanes belonging to phase p_i , approaches the maximum vehicle detection capacity of that lane at the time it is selected to be TP , the associated k_i will increment automatically. This incrementation serves to allocate passage time for potential queuing vehicles outside the field of view, thereby enhancing the efficiency of traffic flow.

Additionally, this scheme incorporates an interruption mechanism that terminates the execution of current phase before its allocated time is depleted and transitions directly to the next phase decision process when there are no vehicles remaining on the lanes belonging to that phase. This mechanism is employed to prevent the occurrence of idle periods, thereby enhancing the efficiency of traffic flow. Concurrently, the appearance of an interruption suggests that the peak for that phase has likely concluded, hence its corresponding k_i is reset to its original value.

B. DRL Based Signal Phase Decision and Control Algorithm

The traffic signal phase decision and control problem studied in this part can be constructed as a MDP model. Consequently, this system employs PPO algorithm as the reinforcement learning scheduling algorithm. The control process and network diagram are depicted in Fig 4.

As shown in Fig. 4, the intersection makes decisions, denoted as actions, based on the observation of the current traffic state values. The decision action is composed of the signal phase ID $phase_{id}$ and the corresponding phase duration $phase_{time}$. Considering the strong correlation between $phase_{id}$ and $phase_{time}$, we adopt a feature cascading method for feature fusion of them. To avoid the issue of local optima, we introduce random noise in the feature input by using Noisy-Dense as the initial feature extraction layer. After executing an action, the system receives the immediate state from the environment and subsequently calculates the reward value *Reward* for the action and the new state data *state* for the next time step. The policy is continuously optimized through the iterative process of the steps mentioned above. Subsequently, this section will provide a

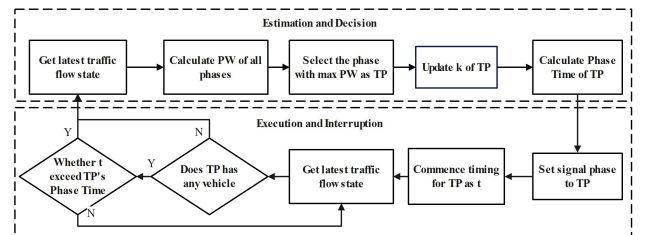


Figure 3. Process of Logical Rule Based Algorithm

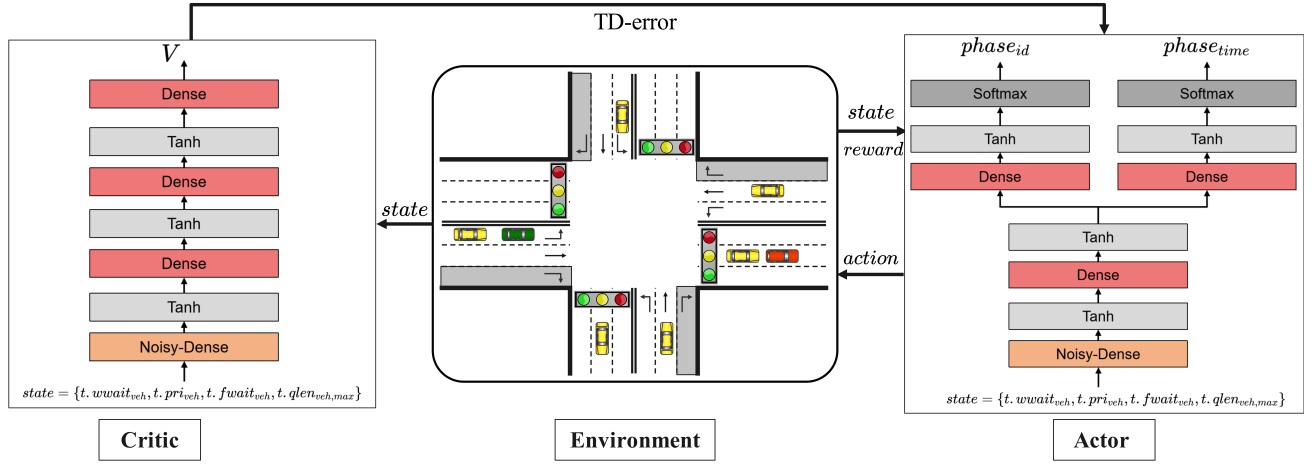


Figure 4. Control Process and Network of DRL Based Signal Phase Decision and Control Algorithm

detailed definition and description of the agent's observation space, action space, and the reward mechanism provided by the environment within this context.

State: The state space of this algorithm encompasses four attributes for each road entering the intersection, which pertain to the TTO of straight type and left type. Specifically, these four attributes include $t.wwait_{veh}$, $t.pri_{veh}$, $t.fwait_{veh}$, and $t.qlen_{veh,max}$.

Action: The action space of this algorithm is comprised of two components: phase actions and phase duration actions. The former is contingent upon the phase scheme in use, while the latter is determined by the intersection's maximum and minimum phase durations.

Reward: The reward calculation method of this algorithm is designed as eq. 15, which consists of the overall waiting time, the μ_4 -th power of the maximum waiting time, and the priority attributes along with the queue length attribute of each TTO. $t.pri_{veh} * t.qlen_{veh,max}$ represent the product of the priority attributes and its queue length attribute. This ensures that when a vehicle associated with a particular TTO is in motion and its waiting queue length attribute is zero, it can receive a larger Reward value, thereby hastening the convergence of the model. Parameters including μ_1 , μ_2 , μ_3 are used to adjust the weights of each item. μ_4 is used to increase the punishment value for waiting too long. The notation $r.ts$ denotes the set of TTOs belonging to road r .

$$Reward = \sum_{r \in j.rs} \sum_{t \in r.ts} (-\mu_1 * t.pri_{veh} * t.qlen_{veh,max} - \mu_2 * t.wwait_{veh} - \mu_3 * t.fwait_{veh}^{\mu_4}) \quad (15)$$

VI. PROTOTYPE REALIZATION AND EXPERIMENTAL VERIFICATION

To validate the adaptive traffic signal phase and timing decision and control mechanism based on multi-mode traffic flow estimation proposed in this paper, we have constructed a prototype system and integrated simulation software such as SUMO and CARLA to develop a semi-physical simulation environment. An analysis of the experimental

results has been performed, confirming the effectiveness of the various mechanisms we have proposed.

A. Prototype System Construction

To assess the practicality of the proposed scheme, we have designed and constructed a prototype system utilizing a complete set of hardware and software configurations. On the software level, we have developed a software system consisting of three modules to meet the verification requirements: the signal phase decision and control module, the traffic flow state estimation module, and the co-simulation environment module. Specifically, the signal phase decision and control module implements the signal phase decision algorithms described in section 5; the traffic flow state estimation module realizes the multi-mode traffic flow perception mechanism detailed in section 4; the co-simulation environment module employs the SUMO-CARLA Co-Simulation Environment as the simulation platform and utilizes the TraCI interface as the interaction tool with the simulation environment. Through this interface, we facilitate the transmission of high-priority vehicle information to the traffic situation perception module and convey the received phase decision results to the simulation environment. A standard crossroad intersection is set as the simulation scenario. The software architecture is depicted in Fig. 5.

On the hardware level, in order to meet the computational requirements of the estimation and decision-making processes, we have employed the Nvidia Jetson AGX Orin (275 TOPS) as the core device, complemented by a variety of other components such as a high-performance Power over Ethernet (36Gbps) switch and high-precision cameras (4K) to construct a comprehensive hardware system

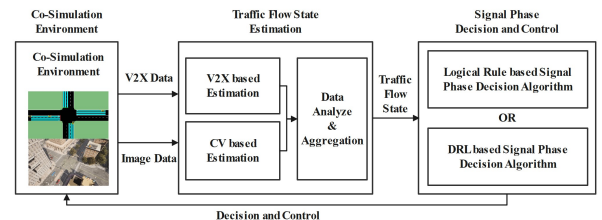


Figure 5. Software Architecture of Prototype System

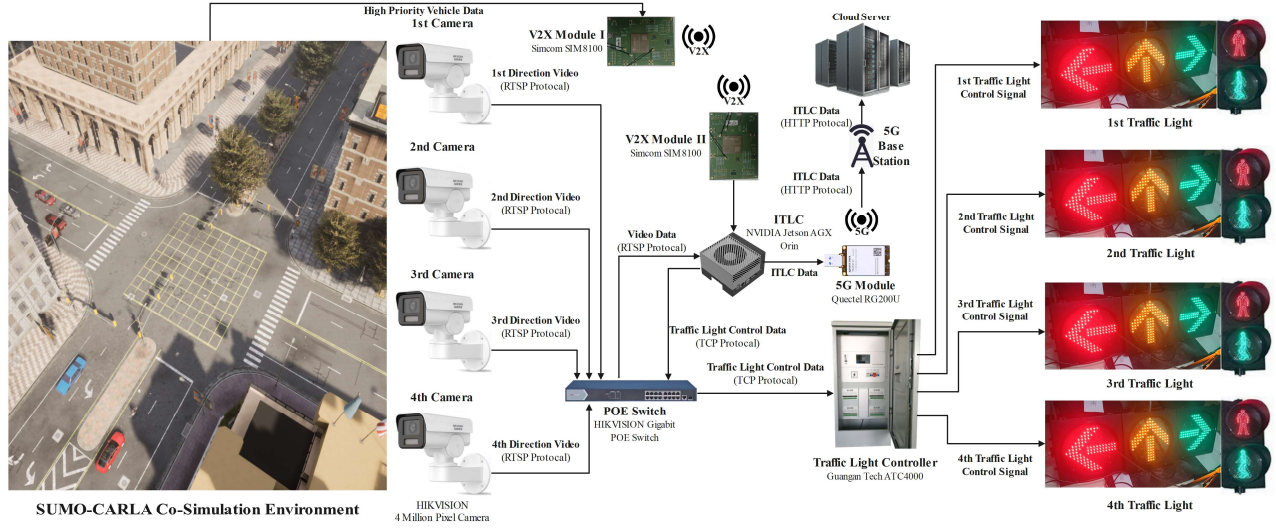


Figure 6. Hardware Architecture of Prototype System

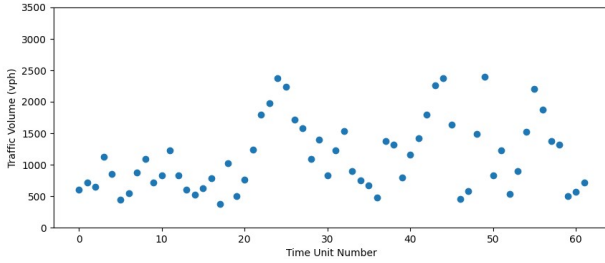


Figure 7. Basic Traffic Volume of Each Time Unit

capable of fulfilling all conditions necessary for the validation of the proposed mechanisms. To more authentically simulate real-world traffic scenarios, we utilize cameras and V2X modules for data transmission and acquisition. The hardware architecture is illustrated in Fig. 6.

The traffic flow utilized in this experiment is based on a unit of 400 seconds, with a total duration of 28,400 seconds. The fundamental traffic volume for each time unit is depicted in Fig. 7.

To simulate more scenarios, the specific traffic volume data has been slightly adjusted from the base vehicle flow. This includes a random distribution treatment for the vehicle flow in each direction on each lane. The number of high-priority vehicles is approximately one-tenth that of low-priority vehicles, yet it has also been adjusted to enrich the scenario.

The experimental simulation scenario features a standard crossroad intersection, with each connecting road consisting of three approach lanes and three departure lanes, which is illustrated in Fig. 8.

B. Experimental Data Analysis

Utilizing the constructed prototype system, we conducted tests on the proposed mechanisms. The experimental data encompasses three categories: the algorithm based on Deep Reinforcement Learning (DRL)

proposed in this paper, the algorithm based on logical rules, the self-organizing method which was proposed in [17] and the fixed-time control (we set 30s to switch phases sequentially) used for comparative purposes. The specific experimental data is depicted in Fig. 9.

From Fig 8, it can be observed that the fixed-time method is unable to adjust the timing intelligently based on traffic flow state, resulting in significantly longer travel time and waiting time compared to the other three types of algorithms. Additionally, logical rule-based algorithm we proposed have respectively compressed the travel times for low-priority vehicles by 30.74%, and 28.67% for high-priority vehicles, waiting times for low-priority vehicles by 39.39%, and 44.46% for high-priority vehicles. Meanwhile, it also surpassed self-organizing method across all metrics. On the other hand, the DRL algorithm has achieved further improvements compared to the logical rule-based algorithm, with a 4% compressed in travel times and a 7% compressed in waiting times. And the DRL algorithm outperforms the logical rule-based algorithm in specific metrics for both high-priority and low-priority vehicles.

Furthermore, it can be noted that during the initial phase of the simulation experiment, when the traffic volume is relatively low, the performance of the algorithm based on logical rules is comparable to that of the algorithm based on DRL, and even slightly superior in some cases. However, as

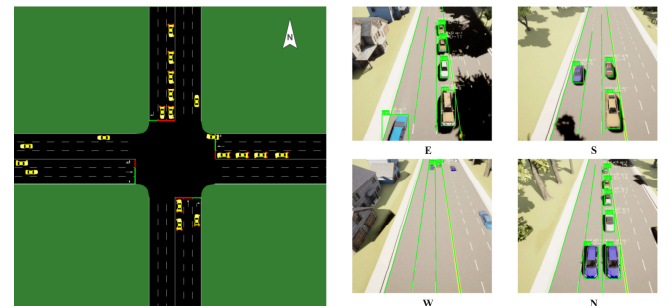


Figure 8. Simulation Scenario

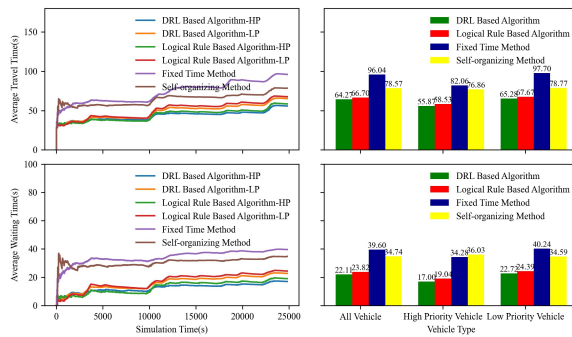


Figure 9. Experiment Data

the traffic volume increases, the DRL-based algorithm is more adept at uncovering the relationship between traffic situation data and optimal decision-making, thus yielding better scheduling outcomes. Additionally, the average waiting times for both types of vehicles are reduced by 10.71% and 6.85%, respectively.

Concurrently, to substantiate the efficacy of the methodologies proposed within this paper, we intend to implement a real-world test by deploying them into the traffic system of a city by the end of the year.

VII. CONCLUSIONS

In this study, we modeled networked vehicles with service attribute, roads, lanes, turn type objects, and intersections. Then, we proposed an adaptive traffic signal phase decision and control framework based on multi-mode traffic flow state estimation. Based on this framework, we researched and proposed a multi-mode, dynamic estimation method for traffic flow situations that integrates vision, communication, and multi-priority. We also developed a signal phase decision and control method that combines logical rules and deep reinforcement learning to effectively handle various intersection scenarios. Finally, we constructed a prototype system base on our framework and conducted experiments in semi-physical simulation environment. The results demonstrated the signal phase decision and control method employed by the proposed framework can reduce the average travel and waiting times of low-priority vehicles by up to 33.18% and 43.54%, respectively, and can reduce the travel and waiting times of high-priority vehicles by 31.92% and 50.41%, respectively. The framework presented in this paper can significantly improve the efficiency of vehicular traffic flow.

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